Supplementary information

Matched nulls reveal specificity limits of metabolic pathway scores

This supplement accompanies the main manuscript. It distinguishes frozen primary analyses, adaptive analyses, locked external replication, post-result robustness analyses and supporting evidence. Machine-readable tables remain authoritative where rounded values are shown here.

Supplementary Methods

Evidence roles and analysis chronology

The ST002081 coarse-family benchmark was frozen before execution. Its structural-descriptor follow-up was specified after the coarse benchmark failed its matched-random specificity gate and is therefore adaptive. ST000818 was selected from a pre-existing public-study registry using only metadata and feature names; its population-held-out protocol was frozen before the abundance matrix was retrieved. The human and CCLE parameter grids were declared after their respective primary results and are robustness analyses, not prospective confirmation. Earlier genetics, constraint, isotope, temporal and interaction analyses are supporting domains and were not pooled with the primary observations.

Matched-null rationale

The no-information population-mean baseline tests whether a representation predicts at all. It does not test whether the declared biological labels matter. The coarse benchmark therefore used random groups preserving family sizes. The lipid-structure benchmark randomized the binary feature-by-descriptor graph while preserving every feature degree and every descriptor degree. The CCLE benchmark used random metabolite, GPR and interaction feature sets matched to each target's feature counts and evaluated with the same lineage folds. These nulls address effective representation complexity; they do not prove causal mechanism.

Group-isolated validation

All samples from a ST002081 participant were held out together. All samples from a ST000818 population category were held out together. All cell lines from a CCLE lineage were held out together in two repeats of five deterministic size-balanced folds. Imputation, scaling, feature summaries and ridge models were fitted using training observations only.

Statistical units

Human confidence intervals used paired resampling of the outer biological groups: participants in ST002081 and population categories in ST000818. The primary analyses used 5,000 bootstrap draws; the post-result grid used 2,000 draws per setting. CCLE comparisons used 5,000 paired bootstrap draws over prediction targets. The stochastic CCLE null ensemble used 2,000 target-bootstrap draws per random seed and 5,000 for the expected-null contrast. These intervals quantify variation across the declared resampling units. They do not make correlated metabolites or reaction neighborhoods independent.

Mapping and source integrity

The feature maps retain every accepted and rejected row and a reason for exclusion. CCLE mappings record assay names, stable identifiers, HumanGEM metabolites and compartments, GPR eligibility, and the exact feature set used for each target. Source and output manifests contain SHA-256 checksums. The publication synthesis verified all 15 contributing artifact manifests.

Supplementary Table 1 | Cohorts and benchmark roles

Resource	Analysis unit	Retained data	Holdout unit	Evidential role
ST002081	Longitudinal human plasma lipidomics	1,539 samples; 112 people; 493 eligible lipids	Participant	Frozen coarse benchmark; adaptive structural analysis
ST000818 / AN001299	Human blood lipidomics	450 samples; 15 population categories; 255 eligible lipids	Population category	Protocol-locked external structural replication
CCLE	Cancer-cell-line metabolomics and RNA	913 cell lines; 76 mapped assay metabolites; 60 primary targets	Lineage	Reaction-neighborhood prediction and transcript ablation
METSIM summary results	Human mQTL states	71 eligible gene–metabolite states	Not re-fitted as individual data	Positive-control topology audit
NSCLC public data	Cell-line abundance, isotope and molecular state	Source-specific eligible observations	Source-defined groups	External falsification of stronger mechanistic claims
HumanGEM / Human1	Curated human metabolic network	Reaction, metabolite and GPR records	Not applicable	Topology and model-relative constraint evidence

Supplementary Table 2 | Primary benchmark contrasts

Benchmark	Biological representation	Matched null	Out-of-group unit	Primary inference
ST002081 coarse	Lipid-family median scores	Random groups preserving family sizes	Participant	Predictive compression, but no family specificity
ST002081 structural	Lipid class, carbon, unsaturation and chain descriptors	Fixed feature and descriptor degree graph	Participant	Adaptive structural specificity
ST000818 structural	Same parser and representation contract	Fixed feature and descriptor degree graph	Population category	External structural replication
CCLE topology	Direct HumanGEM neighbors plus reaction-linked GPR features	Dimension-matched random metabolites, GPRs and products	Cancer lineage	Reaction topology has predictive specificity
CCLE transcript ablation	Direct neighbors plus additive or interaction GPR terms	Nested direct-neighbor/additive models	Cancer lineage	Additive context helps compact prediction; interactions do not improve it generally

Supplementary Table 3 | Human structural sensitivity grid

All entries are structural-minus-degree-null improvements. Parentheses are paired 95% bootstrap intervals. All randomized graphs preserved exact row and column degrees.

Cohort	Setting	Descriptors	Structural RMSE	Null RMSE	RMSE improvement (95% CI)	Precision improvement (95% CI)
ST002081	Primary	95	0.309	0.411	0.104 (0.098, 0.109)	0.133 (0.119, 0.148)
ST002081	Ridge alpha 1	95	0.311	0.408	0.102 (0.097, 0.108)	0.142 (0.129, 0.155)
ST002081	Ridge alpha 100	95	0.326	0.432	0.105 (0.099, 0.112)	0.139 (0.125, 0.153)
ST002081	Descriptor minimum 3	112	0.290	0.375	0.089 (0.084, 0.093)	0.108 (0.095, 0.120)
ST002081	Descriptor minimum 10	69	0.343	0.448	0.109 (0.103, 0.116)	0.179 (0.161, 0.195)
ST002081	Maximum prevalence 0.80	95	0.309	0.407	0.101 (0.095, 0.106)	0.133 (0.119, 0.146)
ST002081	Maximum prevalence 0.95	95	0.309	0.410	0.103 (0.097, 0.110)	0.140 (0.127, 0.153)
ST002081	Five swaps per edge	95	0.309	0.404	0.097 (0.092, 0.102)	0.127 (0.113, 0.140)
ST002081	Twenty-five swaps per edge	95	0.309	0.399	0.093 (0.088, 0.099)	0.126 (0.114, 0.139)
ST000818	Primary	81	1.237	1.790	0.563 (0.057, 1.006)	0.103 (0.086, 0.121)
ST000818	Ridge alpha 1	81	1.288	1.788	0.511 (0.062, 0.903)	0.100 (0.084, 0.117)
ST000818	Ridge alpha 100	81	1.253	1.786	0.543 (0.041, 0.969)	0.074 (0.058, 0.097)
ST000818	Descriptor minimum 3	112	1.192	1.682	0.497 (0.050, 0.882)	0.073 (0.054, 0.092)
ST000818	Descriptor minimum 10	41	1.764	1.821	0.056 (0.042, 0.120)	0.097 (0.081, 0.114)
ST000818	Maximum prevalence 0.80	81	1.237	1.767	0.541 (0.062, 0.960)	0.102 (0.085, 0.121)
ST000818	Maximum prevalence 0.95	81	1.237	1.787	0.561 (0.068, 0.993)	0.102 (0.086, 0.119)
ST000818	Five swaps per edge	81	1.237	1.794	0.569 (0.065, 1.006)	0.102 (0.082, 0.125)
ST000818	Twenty-five swaps per edge	81	1.237	1.773	0.547 (0.057, 0.975)	0.093 (0.071, 0.113)

Decision: all nine settings passed both endpoints in each cohort. Because this grid was designed after the primary results, the decision is `HUMAN_STRUCTURAL_SENSITIVITY_ROBUST`, not independent confirmation.

The graph-quality diagnostic regenerated 200 panel-specific nulls across the two cohorts. All row and column degrees were exact, every panel contained 20 unique nulls, minimum observed-edge replacement was 0.771 and maximum pairwise null Jaccard was 0.131. This demonstrates graph movement and diversity but not perfectly uniform sampling from every graph with the same degrees.

Supplementary Table 4 | CCLE reaction-topology sensitivity grid

The improvement is random-factorized RMSE minus reaction-factorized RMSE, in training-fold outcome standard-deviation units. Positive values favor topology. The final column is interaction-model RMSE improvement over the additive GPR model; negative values favor the additive model.

Setting	Targets	Signatures	Factorized RMSE	Median target R^2	Improvement vs random (95% CI)	Improvement vs additive (95% CI)
Primary	60	169	0.797	0.241	0.124 (0.084, 0.166)	−0.009 (−0.015, −0.004)
Maximum 4 metabolites/reaction	49	75	0.864	0.193	0.084 (0.028, 0.135)	−0.008 (−0.014, −0.002)
Maximum 6 metabolites/reaction	59	154	0.820	0.236	0.099 (0.060, 0.140)	−0.009 (−0.015, −0.004)
Maximum 12 metabolites/reaction	60	198	0.795	0.253	0.131 (0.088, 0.176)	−0.011 (−0.016, −0.005)
Unbounded reaction size	60	202	0.795	0.253	0.125 (0.080, 0.171)	−0.011 (−0.017, −0.006)
Include transport	63	192	0.709	0.469	0.219 (0.169, 0.273)	−0.014 (−0.020, −0.009)
Ridge alpha 1	60	169	0.799	0.237	0.124 (0.083, 0.166)	−0.010 (−0.016, −0.005)
Ridge alpha 100	60	169	0.794	0.252	0.117 (0.079, 0.157)	−0.006 (−0.010, −0.002)
GPR mean	60	169	0.797	0.241	0.123 (0.086, 0.166)	−0.009 (−0.014, −0.004)
GPR sum	60	169	0.797	0.241	0.123 (0.083, 0.167)	−0.009 (−0.014, −0.004)

Decision: reaction topology beat its matched random representation in all ten settings. Transcript interactions failed to beat additive GPR terms in all ten settings. The grid is post-result robustness evidence.

Supplementary Table 5 | CCLE stochastic random-feature null ensemble

The biological feature set and lineage folds were fixed. Each row independently regenerated the dimension-matched random features. Effects are random-minus-topology equal-lineage RMSE.

Random seed	Random RMSE	Improvement (95% target-bootstrap CI)
20262000	0.927	0.130 (0.082, 0.181)
20262001	0.924	0.127 (0.085, 0.171)
20262002	0.902	0.105 (0.055, 0.157)
20262003	0.925	0.128 (0.080, 0.172)
20262004	0.936	0.139 (0.098, 0.185)
20262005	0.921	0.124 (0.071, 0.176)
20262006	0.939	0.142 (0.091, 0.195)
20262007	0.947	0.150 (0.106, 0.195)
20262008	0.922	0.126 (0.074, 0.173)
20262009	0.919	0.122 (0.080, 0.168)
20262010	0.930	0.134 (0.087, 0.184)
20262011	0.924	0.127 (0.085, 0.177)
20262012	0.939	0.143 (0.102, 0.189)
20262013	0.920	0.123 (0.081, 0.168)
20262014	0.926	0.129 (0.082, 0.179)
20262015	0.931	0.134 (0.091, 0.182)

Random seed	Random RMSE	Improvement (95% target-bootstrap CI)
20262016	0.951	0.154 (0.092, 0.232)
20262017	0.970	0.173 (0.111, 0.240)
20262018	0.925	0.128 (0.076, 0.183)
20262019	0.941	0.144 (0.099, 0.196)
Expected null over 20 seeds	0.931	0.134 (0.093, 0.177)

All 20 per-seed intervals excluded zero. The biological-factorized RMSE remained exactly 0.797 SD across seeds. This post-result ensemble addresses stochastic null selection, not external cohort generalization.

Supplementary Table 6 | Property-matched null and subsystem dependence audit

The hard null matched random metabolites on direct-network degree and assay coverage and random GPR signatures on candidate-network degree. It did not use target values or target correlations.

Audit	Units	Effect or balance	95% interval / gate
Property-matched expected null	20 seeds; 60 targets	0.137 SD	0.097–0.181
Property-matched seed range	20 seeds	0.116–0.149 SD	Every per-seed lower bound >0
Metabolite log-degree imbalance	20 seeds	0.055 mean absolute difference	Threshold ≤0.35
Metabolite assay-coverage imbalance	20 seeds	0.000 mean absolute difference	Threshold ≤0.02
GPR log-degree imbalance	20 seeds	0.009 mean absolute difference	Threshold ≤0.35
HumanGEM subsystem-cluster bootstrap	20 clusters; 60 targets	0.137 SD	0.086–0.194
Leave-one-subsystem-out	20 exclusions	Minimum 0.120 SD	Every effect >0

The subsystem audit assigns each target to its most frequent eligible HumanGEM subsystem and is a dependence sensitivity, not proof that pathways or targets are independent.

Supplementary Table 7 | Mapping and eligibility audit

Domain	Total	Accepted	Rejected	Acceptance fraction
ST000818 lipid features	260	255	5	98.1%
ST002081 lipid features	845	493	352	58.3%
CCLE assay metabolites	225	76	149	33.8%
CCLE GPR signatures	195	169	26	86.7%
CCLE primary prediction targets	60	60	0	100.0%

The CCLE conclusion is explicitly limited to the mapped, reaction-accessible assay subset. Rejected rows are retained in `ccle-metabolite-map.csv`; no inferred identity is assigned to an unmapped assay feature.

Supplementary Table 8 | Machine-readable claim ledger

Claim	Decision	Basis
Compact lipid-family scores reconstruct hidden human markers	Supported versus population mean	Participant-bootstrap RMSE interval above zero
Broad lipid-family scores are pathway-specific	Not supported	Size-matched random-group gate failed
Structure-resolved lipid scores beat a fixed-degree null	Adaptive same-cohort support	Participant-paired RMSE and precision intervals above zero
Structure-resolved scores replicate in a separate human cohort	External population-held-out support	ST000818 passed all 20 graph nulls
Direct HumanGEM/GPR features beat size-matched random features	Supported in CCLE	Lineage-isolated target-bootstrap comparison
Transcript interactions improve the general direct-network model	Not supported	Interaction model did not beat direct-metabolite or additive models

Claim	Decision	Basis
Selected transcript-conditioned coupling can exist	Supported in model system	CDA–cytidine–uridine state passed the declared screen and checks
The evidence validates same-person genomic personalization	Not tested	No cohort contains the required participant-level modalities
The model estimates physiological flux or treatment response	Not supported	Constraint evidence is model-relative; external challenges are null
Human structural results survive declared settings	Post-result support	Nine of nine settings pass in each cohort
CCLE reaction topology survives declared settings	Post-result support	Ten of ten settings pass the matched-random gate
CCLE topology is stable across random feature draws	Post-result support	Twenty of twenty seed intervals pass; expected-null lower bound 0.093 SD
CCLE topology beats degree-and-coverage-matched features	Post-result support	Twenty of twenty hard-null intervals pass; expected-null lower bound 0.097 SD
CCLE topology is not concentrated in one subsystem	Post-result support	Twenty-cluster lower bound 0.086 SD; all leave-one-subsystem effects pass
Transcript interactions become generally useful in sensitivity analysis	Not supported	Zero of ten settings beat the additive model
Analysis-facing mappings and exclusions are auditable	Resource supported	Complete accepted/rejected tables and verified checksums

Supplementary Table 9 | Evidence domains not pooled with primary observations

Domain	Result	Adjudication
METSIM–HumanGEM topology	26 of 71 direct recoveries; plus-one empirical $P=9.9999\times10^{-6}$	Supportive positive-control audit; nominations are knowledge based
CCLE reaction-resolved interaction	One FDR-significant state among 284 tests	Candidate-specific model-system evidence
UKB/BBJ summary-statistic scan	One strict replicated row; no novel mutable-context rows	Limited
NSCLC isotope challenge	No expression or oncogenotype FDR signal	Null external challenge
HumanGEM constraint ensemble	Three stable and two unstable configured candidates in 1,000 draws	Model-relative only
ST002081 temporal direction	Forward $R^2=0.452$; reverse $R^2=0.459$	Directional prediction not supported

Supplementary Figure 1 | Factorized benchmark summary

The publication-synthesis figure compares the principal matched-null effects across evidence domains. Effect definitions differ by benchmark and are labeled in the source data; they must not be interpreted as a pooled meta-analytic effect. Source: `artifacts/factorized-pathway-publication/factorized-pathway-benchmarks.png`.

Supplementary Data index

File	Contents
`artifacts/pathway-score-human-sensitivity/sensitivity-grid.csv`	Complete 18-row human parameter grid with paired intervals and degree checks
`artifacts/pathway-score-human-sensitivity/sensitivity-summary.csv`	Cohort-level grid adjudication
`artifacts/pathway-score-human-graph-mixing/null-distance-audit.csv`	Complete 200-row graph movement and degree audit
`artifacts/pathway-score-human-graph-mixing/mask-diversity-summary.csv`	Per-panel uniqueness and pairwise null Jaccard summaries
`artifacts/pathway-score-ccle-sensitivity/setting-summary.csv`	Complete ten-setting CCLE adjudication
`artifacts/pathway-score-ccle-sensitivity/model-metrics-grid.csv`	Metrics for all six models in every CCLE setting
`artifacts/pathway-score-ccle-null-ensemble/seed-summary.csv`	Complete 20-seed stochastic-null results
`artifacts/pathway-score-ccle-null-ensemble/target-null-ensemble.csv`	Target-level random and biological RMSE for every seed
`artifacts/pathway-score-ccle-property-matched-null/seed-summary.csv`	Degree- and coverage-matched null results and balance metrics
`artifacts/pathway-score-ccle-property-matched-null/target-null-ensemble.csv`	Hard-null target-level RMSE values
`artifacts/pathway-score-ccle-reaction-cluster/target-effects-with-subsystem.csv`	Deterministic target-to-subsystem assignments and effects
`artifacts/pathway-score-ccle-reaction-cluster/leave-one-subsystem-out.csv`	Complete leave-one-subsystem-out effects

File	Contents
`artifacts/pathway-score-publication-figures/figure-source-index.csv`	Source path and checksum for every main-figure input
`artifacts/pathway-score-publication-figures/manifest.json`	Figure implementation, configuration and output checksums
`artifacts/pathway-score-mapping-supplement/human-lipid-feature-map.csv`	Human feature eligibility, descriptors and hidden panels
`artifacts/pathway-score-mapping-supplement/ccle-metabolite-map.csv`	Accepted and rejected CCLE assay-to-HumanGEM mappings
`artifacts/pathway-score-mapping-supplement/ccle-gpr-map.csv`	GPR signatures and eligibility
`artifacts/pathway-score-mapping-supplement/ccle-target-feature-sets.csv`	Exact target-level feature sets
`artifacts/pathway-score-mapping-supplement/mapping-qc-summary.csv`	Mapping totals
`artifacts/factorized-pathway-publication/evidence-domain-summary.csv`	Evidence roles and adjudications
`artifacts/factorized-pathway-publication/claim-ledger.csv`	Machine-readable claim decisions and bases
`artifacts/factorized-pathway-publication/manifest.json`	Synthesis decision, source manifests and output checksums

Supplementary boundary statement

The analyses qualify representations for hidden-measurement reconstruction and candidate prioritization. They do not establish pathway activity, reaction direction, physiological flux, causal genotype effects, diagnosis, treatment response, drug efficacy or same-person genomic personalization.